Basics in Mathematical Physics

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1 Complex Analysis

1.1 Analytic Functions

An analytic function is a complex function that is differentiable at every point in its domain. If a function is differentiable within a region, then we say its analytic throughout the region. There are three nice properties of analytic functions:

1. Analytic functions are infinitely differentiable.
2. The contour integral of an analytic function over a closed curve is zero (Cauchy's theorem).
3. Analytic continuiation: If two analytic functions agree on a set of point that has an accumulation point, then they are equal on the entire domain.

Another condition for a function to be analytic is that it must satisfy the Cauchy-Riemann equations. If $f(z) = u(x, y) + iv(x, y)$, where $z = x + iy$, then the Cauchy-Riemann equations are given by:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}. \tag{1}$$

1.2 Cauchy's Integral Theorem

Theorem 1.1 (Cauchy's Integral Theorem)**.** *If f is analytic in a simply connected domain D, then for any closed contour C in D, we have:*

$$\oint_C f(z)\,dz = 0. \tag{2}$$

Proof. We first expand $f(z)$ in terms of its real and imaginary parts, then the contour integral can be expressed as:

$$\oint_C f(z)\,dz = \oint_C (u + iv)(dx + idy) = \oint_C (udx - vdy) + i \oint_C (vdx + udy). \tag{3}$$

By Green's theorem, we can convert the line integrals into double integrals over the region R enclosed by C:

$$\oint_C (udx - vdy) = \iint_R \left(\frac{\partial(-v)}{\partial x} - \frac{\partial u}{\partial y}\right) dA, \quad \oint_C (vdx + udy) = \iint_R \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right) dA. \tag{4}$$

Since f is analytic, it satisfies the Cauchy-Riemann equations,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}. \tag{5}$$

Substituting these into the double integrals, we find that both integrals evaluate to zero:

$$\oint_C (udx - vdy) = 0, \quad \oint_C (vdx + udy) = 0. \tag{6}$$

Therefore, we conclude that:

$$\oint_C f(z)\,dz = 0. \tag{7}$$

□

1.3 Cauchy's Integral Theorem

Theorem 1.2 (Cauchy's Integral Formula)**.** *If f is analytic in a simply connected domain D and C is a positively orirented simple closed contour in D, then for any point z_0 inside C, we have:*

$$f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz. \tag{8}$$

Proof. For an arbitraryly small $\epsilon > 0$, we can find a δ such that for $|z - z_0| < \delta$

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \lambda(z)(z - z_0), \tag{9}$$

where $|\lambda(z)| < \epsilon$ (the definition of $f(z)$ being analytic at z_0). Deforming the contour to be the circle of radius δ about z_0 we find

$$\begin{aligned} \oint_C \frac{f(z)}{z - z_0} dz &= f(z_0) \oint_C \frac{1}{z - z_0} dz + f'(z_0) \oint_C dz + \oint_C \lambda(z) dz \\ &= i2\pi f(z_0) + \oint_C \lambda(z) dz \end{aligned} \tag{10}$$

As

$$\left| \oint_C \lambda(z) dz \right| \leq \oint_C |\lambda(z)||dz| < \epsilon 2\pi\delta, \tag{11}$$

we have

$$\left| \oint_C \frac{f(z)}{z - z_0} dz - 2\pi i f(z_0) \right| < \epsilon 2\pi\delta. \tag{12}$$

Since ϵ is arbitrarily small, we have

$$\oint_C \frac{f(z)}{z - z_0} dz = 2\pi i f(z_0). \tag{13}$$

□

1.4 Derivatives from Cauchy's Integral Theorem

Takes derivative of the Cauchy's Integral Formula with respect to z_0, we have

$$f'(z_0) = \frac{1}{2\pi i} \frac{d}{dz_0} \oint_C \frac{f(z)}{z - z_0} dz = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^2} dz. \tag{14}$$

Then more genrally we will have the nth order derivative of f at z_0 as

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz. \tag{15}$$

1.5 Liouville's Theorem

Theorem 1.3 (Liouville's Theorem)**.** *If f is entire and bounded, then f is constant.*

Proof. If f is entire and bounded, then there exist a constant M such that $|f(z)| \leq M$ in the entire complex plane. By Cauchy's integral formula for the first derivative, we have:

$$f'(z) = \frac{1}{2\pi i} \oint_C \frac{f(w)}{(w - z)^2} dw, \tag{16}$$

where C is a circle of radius R centered at z. Taking the modulus and using the bound on f, we get:

$$|f'(z)| \leq \frac{1}{2\pi} \oint_C \frac{|f(w)|}{|w - z|^2} |dw| \leq \frac{1}{2\pi} \cdot M \cdot \frac{2\pi R}{R^2} = \frac{M}{R}. \tag{17}$$

As $R \to \infty$, we have $|f'(z)| \to 0$, which means $f'(z) = 0$ for all z. Therefore, f is constant. □

1.6 Laurent Expansion

This is a very important theorem in complex analysis to express an analytic funcition in the form of power series. The Laurent expansion is a representation of a complex function as a series that includes both positive and negative powers of the variable. It is particularly useful for functions that have singularities, as it allows us to express the function in a neighborhood of those singularities.

Theorem 1.4 (Laurent Expansion)**.** *If f is analytic in an annulus $r < |z - z_0| < R$, then f has a unique representation as a Laurent series:*

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n, \tag{18}$$

where the coefficients are given by:

$$a_n = \frac{1}{2\pi i} \oint_C \frac{f(w)}{(w - z_0)^{n+1}} dw, \tag{19}$$

and C is any simple closed contour in the annulus, oriented positively.

If the function f is analytic throughout the whole disk, then the Laurent series reduces to a Taylor series. If f has a singularity at z_0, then the Laurent series will have negative powers of $(z - z_0)$. The principal part of the Laurent series is the sum of the terms with negative powers, and it captures the behavior of the function near the singularity.

Proof. If f is analytic in the annulus $r < |z - z_0| < R$, then the Cauchy integral formula gives us:

$$f(z) = \frac{1}{2\pi i} \oint_{C_1} \frac{f(w)}{w - z} dw - \frac{1}{2\pi i} \oint_{C_2} \frac{f(w)}{w - z} dw, \tag{20}$$

where C_1 is a positively oriented contour around z_0 with radius R, and C_2 is a negatively oriented contour around z_0 with radius r. We can expand the integrands in geometric series:

$$\frac{1}{w - z} = \frac{1}{w - z_0 - (z - z_0)} = \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(w - z_0)^{n+1}} \quad \text{for } |z - z_0| < |w - z_0|, \tag{21}$$

For the second term, we have:

$$-\frac{1}{w - z} = -\frac{1}{(z - z_0) - (w - z_0)} = \sum_{m=0}^{\infty} \frac{(w - z_0)^m}{(z - z_0)^{m+1}} \quad \text{for } |z - z_0| > |w - z_0|. \tag{22}$$

Then we substitue $n = -m - 1$ in the second series, we have

$$-\frac{1}{w - z} = \sum_{n=-\infty}^{-1} \frac{(w - z_0)^{-n-1}}{(z - z_0)^{-n}}. \tag{23}$$

Putting them together, we have

$$f(z) = \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \oint_{C_1} \frac{f(w)}{(w - z_0)^{n+1}} dw \right) (z - z_0)^n + \sum_{n=-\infty}^{-1} \left(\frac{1}{2\pi i} \oint_{C_2} \frac{f(w)}{(w - z_0)^{n+1}} dw \right) (z - z_0)^n. \tag{24}$$

Then this is the Laurent expansion of $f(z)$ in the annulus $r < |z - z_0| < R$,

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n, \tag{25}$$

□

1.7 Isolated Singularities

Definition 1.1 (Isolated Singularity)**.** *A point $z_0 \in \mathbb{C}$ is called an isolated singularity of a complex function $f(z)$ if there exist $r > 0$ such that $f(z)$ is analytic in the punctured disk $0 < |z - z_0| < r$, but $f(z)$ is not analytic at z_0.*

Definition 1.2 (Removable Singularity)**.** *A removable singularity is an isolated singularity of a complex function $f(z)$ at a point z_0 such that the limit $\lim_{z\to z_0} f(z)$ exists and is finite. In this case, we can define a new function $\tilde{f}(z)$ that is equal to $f(z)$ for $z \neq z_0$ and is defined to be $\tilde{f}(z_0) = \lim_{z\to z_0} f(z)$. The new function $\tilde{f}(z)$ is then analytic at z_0, effectively "removing" the singularity.*

Definition 1.3 (Pole)**.** *A pole of order m is an isolated singularity of a complex function $f(z)$ at a point z_0 such that the Laurent series expansion of $f(z)$ around z_0 has $a_{-m} \neq 0$ and $a_{-n} = 0$ for $n > m$.*

In particular, the Laurent expansion for a order of m pole of complex function $f(z)$ in the region $0 < |z - z_0| < R$ is given by

$$f(z) = \sum_{k=0}^{\infty} a_k (z - z_0)^k + \frac{a_{-1}}{z - z_0} + \frac{a_{-2}}{(z - z_0)^2} + \cdots + \frac{a_{-m}}{(z - z_0)^m}. \tag{26}$$

Definition 1.4 (Essential Singularity)**.** *An essential singularity is a type of isolated singularity of a complex function where the Laurent expansion of the function around that point contains infinitely many negative power terms.*

Then the Laurent expansion for a complex function $f(z)$ about its essential singularity is

$$f(z) = \sum_{k=0}^{\infty} a_k (z - z_0)^k + \sum_{n=-\infty}^{-1} a_n (z - z_0)^n, \quad a_n \neq 0 \text{ for infinitely many } n \tag{27}$$

1.8 Residue Theorem

Theorem 1.5 (Residue Theorem)**.** *Let f be analytic on and inside a positively oriented simple closed contour C, except at finitely many isolated singularities $z_1, z_2, \ldots, z_n$ inside C. Then the contour integral of f around C is given by:*

$$\oint_C f(z)dz = 2\pi i \sum_{k=1}^{N} Res(f, z_k), \quad Res(f, z_k) := a_{-1}^{(k)}, \tag{28}$$

where $a_{-1}^{(k)}$ is the coefficient of $(z - z_k)^{-1}$ in the Laurent series expansion of f around the singularity z_k.

Proof. For $\rho > 0$ and $n \in \mathbb{Z}$, let C_ρ denote the circle $|z - z_k| = \rho$ traversed counterclockwise. Parameterise $z = z_k + \rho e^{i\theta}$, so $dz = i\rho e^{i\theta}\, d\theta$. Then

$$\oint_{C_\rho} (z - z_k)^n \, dz = i\rho^{\,n+1} \int_0^{2\pi} e^{i(n+1)\theta}\, d\theta = \begin{cases} 2\pi i, & n = -1, \\ 0, & n \neq -1. \end{cases} \tag{29}$$

Let z_k be an isolated singularity of f. Then on $0 < |z - z_k| < R_k$ we have

$$f(z) = \sum_{n=-\infty}^{\infty} a_n^{(k)} (z - z_k)^n. \tag{30}$$

Choose $0 < \rho < R_k$. The series converges uniformly on compact subsets of the annulus, and C_ρ is compact, so the series converges uniformly on C_ρ. In particular, if $S_M = \sum_{|n|\leq M} a_n^{(k)} (z - z_k)^n$ is the partial sum, then $\sup_{C_\rho} |f - S_M| \to 0$ as $M \to \infty$, and hence

$$\left| \oint_{C_\rho} (f - S_M)\, dz \right| \leq 2\pi\rho \sup_{C_\rho} |f - S_M| \xrightarrow{M\to\infty} 0. \tag{31}$$

So the integral and the sum may be exchanged:

$$\oint_{C_\rho} f(z)\,dz = \sum_{n=-\infty}^{\infty} a_n^{(k)} \oint_{C_\rho} (z - z_k)^n\,dz = 2\pi i\, a_{-1}^{(k)}. \tag{32}$$

Choose ρ small enough that the discs $|z - z_k| \leq \rho$ are disjoint, lie inside C, and satisfy $\rho < R_k$. Let C_k be the counterclockwise circle $|z - z_k| = \rho$. Excise these discs from the interior of C and join them to C by thin cuts, each traversed once in each direction. The cut contributions cancel, leaving C counterclockwise and each C_k clockwise as the boundary of a region where f is analytic.

Cauchy's theorem then gives

$$\oint_C f(z)\,dz - \sum_{k=1}^{N} \oint_{C_k} f(z)\,dz = 0. \tag{33}$$

Therefore,

$$\oint_C f(z)\,dz = \sum_{k=1}^{N} \oint_{C_k} f(z)\,dz = 2\pi i \sum_{k=1}^{N} a_{-1}^{(k)} = 2\pi i \sum_{k=1}^{N} \operatorname{Res}(f, z_k). \tag{34}$$

□

To actually evaluate the integral, we need to compute the residue of the function at each singularity. The residue can be computed in serveal ways depending on the type of singularity. For a pole of order m at z_0, we shall consider the function $g(z) = (z - z_0)^m f(z)$, which is analytic at z_0.

1.9 Evaluating Real Integrals Using Residue Theorem

1.9.1 Trigonometry Functions

As we can always express trigonometric functions in terms of complex exponential functions, we can use the residue theorem to evaluate integrals involving trigonometric functions by transforming the integral into a contour integral in the complex plane. For a real integral of the form

$$I = \int_0^{2\pi} f(cos\theta, \sin\theta)d\theta, \tag{35}$$

we can always transform it by using the substitution

$$\begin{aligned} \cos\theta &= \frac{1}{2}(e^{i\theta} + e^{-i\theta}) = \frac{1}{2}\left(z + \frac{1}{z}\right), \\ \sin\theta &= \frac{1}{2i}(e^{i\theta} - e^{-i\theta}) = \frac{1}{2i}\left(z - \frac{1}{z}\right), \\ d\theta &= \frac{dz}{iz}, \quad z = e^{i\theta}. \end{aligned} \tag{36}$$

Then the integral becomes

$$I = \int_{|z|=1} f\left(\frac{1}{2}\left(z + \frac{1}{z}\right), \frac{1}{2i}\left(z - \frac{1}{z}\right)\right)\frac{dz}{iz}. \tag{37}$$

Then, we can evaluate the integral using the residue theorem by finding the residues of the transformed function inside the unit circle.

1.10 Improper Integral from $-\infty$ to ∞

For an improper integral of the form

$$I = \int_{-\infty}^{\infty} f(x)dx, \tag{38}$$

we can evaluate the integral using the residue theorem when the function f is analytic on the real axis and in the upper half plane except at a finite number of poles in the uuper half plane. We can consider a

semicircular contour in the upper half plane that consists of the real axis from $-R$ to R and a semicircle of radius R centered at the origin. As $R \to \infty$, the integral over the semicircle vanishes if $f(z)$ decays sufficiently fast at infinity, and we are left with the integral over the real axis.

Theorem 1.6 (Jordan's Lemma)**.** *Let $f(z)$ be a complex function such that $|f(z)| \leq M/|z|^p$ for some $p > 1$ and all sufficiently large $|z|$. Then, for any real number $a > 0$, we have*

$$\lim_{R\to\infty} \int_{C_R} f(z)e^{iaz}dz = 0, \tag{39}$$

where C_R is the semicircular contour of radius R in the upper half-plane.

Proof. Taking the modulus of the exponential function, we have

$$|e^{iaz}| = |e^{ia(x+iy)}| = |e^{iaR\cos\theta}e^{-aR\sin\theta}| = |e^{iaR\cos\theta}||e^{-aR\sin\theta}| = e^{-aR\sin\theta}. \tag{40}$$

Then we consider the ML inequality for the integral over the semicircular contour C_R:

$$\left|\oint_{C_R} f(z)e^{iaz}dz\right| \leq M(R)R\int_0^\pi e^{-aR\sin\theta}d\theta, \tag{41}$$

as $\sin\theta$ is symmetric about $\theta = \pi/2$, we have the Jordan's inequality on $[0, \pi/2]$:

$$\sin\theta \geq \frac{2\theta}{\pi} \tag{42}$$

Then the bound becomes

$$\int_0^{\pi/2} e^{-aR\sin\theta}d\theta \leq \int_0^{\pi/2} e^{-2aR\theta/\pi}d\theta = \frac{\pi}{2aR}\left(1 - e^{-aR}\right). \tag{43}$$

Combining,

$$\left|\oint_{C_R} f(z)e^{iaz}dz\right| \leq \frac{\pi M(R)}{a}\left(1 - e^{-aR}\right). \tag{44}$$

So then, the only requirment is that $M(R) \xrightarrow{R\to\infty} 0$, which is the condition $p \geq 1$. □

2 Differential Equations